

Instructions. (15 points) Sections 15.3, 15.5, 15.6. Show relevant work for full credit.

(4^{pts}) 1.

Evaluate $\iint_D e^{-(x^2+y^2)} dA$, where D is the region between the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.

Solution:

Convert to polar: $x^2 + y^2 = r^2$, $dA = r dr d\theta$. The region is $1 \leq r \leq 2$, $0 \leq \theta \leq 2\pi$.

$$\int_0^{2\pi} \int_1^2 e^{-r^2} r dr d\theta$$

Inner integral: Let $u = -r^2$, $du = -2r dr$:

$$\int_1^2 r e^{-r^2} dr = \left[-\frac{1}{2} e^{-r^2} \right]_1^2 = -\frac{1}{2} e^{-4} + \frac{1}{2} e^{-1} = \frac{1}{2} (e^{-1} - e^{-4})$$

Outer integral:

$$\int_0^{2\pi} \frac{1}{2} (e^{-1} - e^{-4}) d\theta = \boxed{\pi(e^{-1} - e^{-4})}$$

(6^{pts}) 2.

(a) (3 pts) Find the surface area of the part of the plane $z = 3 + x - 2y$ that lies above the rectangle $0 \leq x \leq 2$, $0 \leq y \leq 1$.

Solution:

$f(x, y) = 3 + x - 2y$, so $f_x = 1$ and $f_y = -2$.

$$\sqrt{f_x^2 + f_y^2 + 1} = \sqrt{1 + 4 + 1} = \sqrt{6}$$

Since the integrand is constant:

$$A = \sqrt{6} \cdot (\text{area of rectangle}) = \sqrt{6} \cdot (2)(1) = \boxed{2\sqrt{6}}$$

(b) (3 pts) Set up, but do not evaluate, the surface area integral for $z = x^2 + y^2$ over the disk $x^2 + y^2 \leq 4$. Write your answer in polar coordinates.

Solution:

$f(x, y) = x^2 + y^2$, so $f_x = 2x$ and $f_y = 2y$.

$$\sqrt{f_x^2 + f_y^2 + 1} = \sqrt{4x^2 + 4y^2 + 1} = \sqrt{4r^2 + 1}$$

$$A = \int_0^{2\pi} \int_0^2 \sqrt{4r^2 + 1} r dr d\theta$$

(Note the extra r from $dA = r dr d\theta$.)

- (3^{pts}) **3.** Evaluate $\iiint_E z \, dV$, where E is the solid bounded by the planes $x = 0$, $y = 0$, $z = 0$, and $x + y + z = 1$.

Solution:

The solid is a tetrahedron with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$.

Limits ($dz \, dy \, dx$): z from 0 to $1 - x - y$; y from 0 to $1 - x$; x from 0 to 1.

$$\int_0^1 \int_0^{1-x} \int_0^{1-x-y} z \, dz \, dy \, dx$$

Inner: $\int_0^{1-x-y} z \, dz = \frac{(1-x-y)^2}{2}$

Middle: $\int_0^{1-x} \frac{(1-x-y)^2}{2} \, dy$. Let $u = 1 - x - y$, $du = -dy$:

$$= \frac{1}{2} \int_0^{1-x} u^2 \, du = \frac{1}{2} \cdot \frac{(1-x)^3}{3} = \frac{(1-x)^3}{6}$$

Outer: $\int_0^1 \frac{(1-x)^3}{6} \, dx = \frac{1}{6} \cdot \frac{(1-x)^4}{-4} \Big|_0^1 = \frac{1}{6} \cdot \frac{1}{4} = \boxed{\frac{1}{24}}$

- (2^{pts}) **4.** Set up, but do not evaluate, a triple integral for the volume of the solid in the first octant bounded by $z = 4 - y^2$ and $x = 3$.

Solution:

The solid is bounded by:

- Bottom: $z = 0$
- Top: $z = 4 - y^2$ (parabolic cylinder)
- Back: $x = 3$
- First octant: $x \geq 0$, $y \geq 0$, $z \geq 0$

The top surface hits $z = 0$ when $y^2 = 4$, so $y = 2$ (first octant).

$$V = \int_0^3 \int_0^2 \int_0^{4-y^2} dz \, dy \, dx$$

(Or equivalently: $V = \int_0^2 \int_0^3 \int_0^{4-y^2} dz \, dx \, dy$.)